

2.10 Organic chemistry

Related topics in Units 4 and 5 will assume knowledge of this material.

Students will be assessed on their ability to:

1 Alcohols

- a give examples of, and recognise, molecules that contain the alcohol functional group
- b demonstrate an understanding of the nomenclature and corresponding structural, displayed and skeletal formulae of alcohols, and classify them as primary, secondary or tertiary
- c describe the following chemistry of alcohols:
 - i combustion
 - ii reaction with sodium
 - iii substitution reactions to form halogenoalkanes, including reaction with PCl_5 and its use as a qualitative test for the presence of the $-\text{OH}$ group
 - iv oxidation using potassium dichromate(VI) in dilute sulfuric acid on primary alcohols to produce aldehydes and carboxylic acids and on secondary alcohols to produce ketones
- d demonstrate an understanding of, and practise, the preparation of an organic liquid (reflux and distillation), e.g. oxidation of alcohols.

2 Halogenoalkanes

- a demonstrate an understanding of the nomenclature and corresponding structural, displayed and skeletal formulae for halogenoalkanes, including the distinction between primary, secondary and tertiary structures
- b interpret given data and observations comparing the reactions and reactivity of primary, secondary and tertiary compounds
- c carry out the preparation of an halogenoalkane from an alcohol and explain why a metal halide and concentrated sulfuric acid should not be used when making a bromoalkane or an iodoalkane

- d describe the typical behaviour of halogenoalkanes. This will be limited to treatment with:
- i aqueous alkali, e.g. KOH (aq)
 - ii alcoholic potassium hydroxide
 - iii water containing dissolved silver nitrate
 - iv alcoholic ammonia
- e carry out the reactions described in 2.10.2d i, ii, iii
- f discuss the uses of halogenoalkanes, e.g. as fire retardants and modern refrigerants.